FACULDADE DE DESIGN,
TECNOLOGIA E COMUNICAÇÃO
DA UNIVERSIDADE EUROPEIA

Universidade Europeia

MASTER'S IN CREATIVE COMPUTING
AND ARTIFICIAL INTELLIGENCE

EEG-Based Remote Car Control with Emotiv EPOC X:

An Open-Source Python Framework and Experimental Evaluation

Dumilde José de Matos

Supervisors:

Prof. Dr. Pamela Teubig
Prof. Dr. João A. Dias

IADE, Faculdade de Design, Tecnologia e Comunicação
da Universidade Europeia

October 2026

Faculdade de Design, Tecnologia e Comunicação
Universidade Europeia

2026

DUMILDE JOSÉ DE MATOS

EEG-BASED REMOTE CAR CONTROL WITH EMOTIV EPOC X: AN OPEN-SOURCE PYTHON FRAMEWORK AND EXPERIMENTAL EVALUATION

MOD-195.IADEV02; 07-07-2023

Faculdade de Design, Tecnologia e Comunicação
Universidade Europeia

2026

DUMILDE JOSÉ DE MATOS

EEG-BASED REMOTE CAR CONTROL WITH EMOTIV EPOC X: AN OPEN-SOURCE PYTHON FRAMEWORK AND EXPERIMENTAL EVALUATION

Dissertação apresentada ao IADE - Faculdade de Design, Tecnologia e Comunicação da Universidade Europeia, para cumprimento dos requisitos necessários à obtenção do grau de Mestre em Creative Computing and Artificial Intelligence realizada sob a orientação científica da Doutora Pamela Teubig, professora assistente do *IADE* e do Doutor João A. Dias, professor do IADE.

MOD-195.IADEV02; 07-07-2023

Acknowledgements *[Text to write.]*

MOD-195.IADEV02; 07-07-2023

Keywords *[Keywords to add.]*

abstract *[Text to write.]*

Glossary

Term *[Definition to write.]*

List of Abbreviations

ABBR *[Meaning to write.]*

Contents

Glossary **1**

List of Abbreviations **2**

List of Figures **6**

List of Tables **7**

1 Introduction **8**
1.1 Background and Context 8
1.2 Motivation 8
1.3 Problem Statement 8
1.4 Research Gap 8
1.5 Research Question 8
1.6 Main Research Objective 8
1.7 Specific Research Objectives 8
1.8 Research Hypotheses 8
1.9 Scope and Limitations 8
1.10 Expected Contributions 8
1.11 Dissertation Structure 8

2 Brain–Computer Interfaces and EEG-Based Control: A Literature Review **9**
2.1 Introduction 9
2.2 Brain–Computer Interfaces 9
2.3 EEG Technology and Brain Signal Acquisition 9
2.4 EEG-Based Control Paradigms 9
2.5 Emotiv EPOC X: Architecture and Capabilities 9
2.6 Emotiv Mental Commands 9
2.7 Consumer-Grade EEG Devices in BCI Applications 9
2.8 Machine Learning Approaches for EEG Classification 9
2.9 EEG-Based Vehicle and Mobile Robot Control 9
2.10 Real-Time BCI Performance Challenges 9
2.10.1 Command Detection and Classification Reliability 10
2.10.2 Response Latency 10
2.10.3 False Activations and Missed Commands 10
2.10.4 Cognitive Workload and User Adaptation 10
2.11 Safety and Reliability in Brain-Controlled Systems 10
2.12 Existing Frameworks and Related Research 10
2.13 Research Gap and Chapter Summary 10

3 Design and Implementation of the Python Framework **11**
3.1 Introduction 11
3.2 System Requirements 11
3.3 Overall System Architecture 11

3.4 Hardware and Software Components . . . 11
3.5 Integration with the Emotiv Cortex API . . . 11
3.6 Mental Commands Acquisition and Processing . . . 11
3.7 Command Mapping and Control Logic . . . 11
3.8 Communication Between the Framework and Remote Car . . . 11
3.9 Real-Time Control Mechanism . . . 11
3.10 Safety Mechanisms and Emergency Stop . . . 11
3.11 Data Logging and Experimental Recording . . . 12
3.12 Software Testing and Validation . . . 12
3.13 Open-Source Design and Reproducibility . . . 12
3.14 Chapter Summary . . . 12

4 Methodology **13**
4.1 Introduction . . . 13
4.2 Research Design and Approach . . . 13
4.3 Experimental Setup and Equipment . . . 13
4.4 Participant and Ethical Considerations . . . 13
4.5 Experimental Environment . . . 13
4.6 Command Training and Calibration . . . 13
4.7 Experimental Tasks and Protocol . . . 13
4.8 Pilot Study . . . 13
4.9 Main Experimental Sessions . . . 13
4.10 Performance Metrics . . . 13
4.10.1 Command Success Rate . . . 13
4.10.2 False Activation Rate . . . 14
4.10.3 Missed Command Rate . . . 14
4.10.4 Command Detection Latency . . . 14
4.10.5 End-to-End Response Time . . . 14
4.10.6 Task Completion Rate . . . 14
4.10.7 Performance Variation Across Sessions . . . 14
4.11 Data Collection and Management . . . 14
4.12 Data Analysis Methods . . . 14
4.13 Optional Investigation of Custom Machine Learning . . . 14
4.14 Validity, Reliability and Reproducibility . . . 14
4.15 Chapter Summary . . . 14

5 Results **15**
5.1 Introduction . . . 15
5.2 Experimental Data Overview . . . 15
5.3 Command Detection Performance . . . 15
5.4 Command Success and Missed Command Rates . . . 15
5.5 False Activation Results . . . 15
5.6 Command Detection and End-to-End Latency . . . 15
5.7 Remote-Car Task Completion Results . . . 15
5.8 Performance Variation Across Experimental Sessions . . . 15
5.9 Optional Custom Machine Learning Results . . . 15

5.10 Summary of Findings 15

6 Discussion **16**
6.1 Introduction 16
6.2 Interpretation of Experimental Results 16
6.3 Evaluation Against Research Objectives 16
6.4 Reliability and Accuracy of Mental Commands 16
6.5 Latency and Real-Time Control Performance 16
6.6 Trade-Offs Between Reliability and Response Time 16
6.7 Implications for Consumer-Grade EEG Applications 16
6.8 Comparison with Previous Research 16
6.9 Limitations of the Framework and Experimental Setup 16
6.10 Future Improvements 16
6.11 Chapter Summary 16

7 Conclusion **17**
7.1 Introduction 17
7.2 Summary of the Research 17
7.3 Main Contributions 17
7.4 Achievement of Research Objectives 17
7.5 Research Limitations 17
7.6 Recommendations for Future Research 17
7.7 Final Remarks 17

References **18**

Annex **19**
A. Experimental Protocol
B. Participant Information and Consent Documents
C. Supplementary Experimental Results
D. System Architecture and Technical Configuration
E. Source Code and Repository Information
F. Experimental Data and Reproduction Instructions

List of Figures

List of Tables

1 Introduction

[Text to write.]

1.1 Background and Context

[Text to write.]

1.2 Motivation

[Text to write.]

1.3 Problem Statement

[Text to write.]

1.4 Research Gap

[Text to write.]

1.5 Research Question

[Text to write.]

1.6 Main Research Objective

[Text to write.]

1.7 Specific Research Objectives

[Text to write.]

1.8 Research Hypotheses

[Text to write.]

1.9 Scope and Limitations

[Text to write.]

1.10 Expected Contributions

[Text to write.]

1.11 Dissertation Structure

[Text to write.]

2 Brain–Computer Interfaces and EEG-Based Control: A Literature Review

[Text to write.]

2.1 Introduction

[Text to write.]

2.2 Brain–Computer Interfaces

[Text to write.]

2.3 EEG Technology and Brain Signal Acquisition

[Text to write.]

2.4 EEG-Based Control Paradigms

[Text to write.]

2.5 Emotiv EPOC X: Architecture and Capabilities

[Text to write.]

2.6 Emotiv Mental Commands

[Text to write.]

2.7 Consumer-Grade EEG Devices in BCI Applications

[Text to write.]

2.8 Machine Learning Approaches for EEG Classification

[Text to write.]

2.9 EEG-Based Vehicle and Mobile Robot Control

[Text to write.]

2.10 Real-Time BCI Performance Challenges

[Text to write.]

2.10.1 Command Detection and Classification Reliability

[Text to write.]

2.10.2 Response Latency

[Text to write.]

2.10.3 False Activations and Missed Commands

[Text to write.]

2.10.4 Cognitive Workload and User Adaptation

[Text to write.]

2.11 Safety and Reliability in Brain-Controlled Systems

[Text to write.]

2.12 Existing Frameworks and Related Research

[Text to write.]

2.13 Research Gap and Chapter Summary

[Text to write.]

3 Design and Implementation of the Python Framework

[Text to write.]

3.1 Introduction

[Text to write.]

3.2 System Requirements

[Text to write.]

3.3 Overall System Architecture

[Text to write.]

3.4 Hardware and Software Components

[Text to write.]

3.5 Integration with the Emotiv Cortex API

[Text to write.]

3.6 Mental Commands Acquisition and Processing

[Text to write.]

3.7 Command Mapping and Control Logic

[Text to write.]

3.8 Communication Between the Framework and Remote Car

[Text to write.]

3.9 Real-Time Control Mechanism

[Text to write.]

3.10 Safety Mechanisms and Emergency Stop

[Text to write.]

3.11 Data Logging and Experimental Recording

[Text to write.]

3.12 Software Testing and Validation

[Text to write.]

3.13 Open-Source Design and Reproducibility

[Text to write.]

3.14 Chapter Summary

[Text to write.]

4 Methodology

[Text to write.]

4.1 Introduction

[Text to write.]

4.2 Research Design and Approach

[Text to write.]

4.3 Experimental Setup and Equipment

[Text to write.]

4.4 Participant and Ethical Considerations

[Text to write.]

4.5 Experimental Environment

[Text to write.]

4.6 Command Training and Calibration

[Text to write.]

4.7 Experimental Tasks and Protocol

[Text to write.]

4.8 Pilot Study

[Text to write.]

4.9 Main Experimental Sessions

[Text to write.]

4.10 Performance Metrics

[Text to write.]

4.10.1 Command Success Rate

[Text to write.]

4.10.2 False Activation Rate

[Text to write.]

4.10.3 Missed Command Rate

[Text to write.]

4.10.4 Command Detection Latency

[Text to write.]

4.10.5 End-to-End Response Time

[Text to write.]

4.10.6 Task Completion Rate

[Text to write.]

4.10.7 Performance Variation Across Sessions

[Text to write.]

4.11 Data Collection and Management

[Text to write.]

4.12 Data Analysis Methods

[Text to write.]

4.13 Optional Investigation of Custom Machine Learning

[Text to write.]

4.14 Validity, Reliability and Reproducibility

[Text to write.]

4.15 Chapter Summary

[Text to write.]

5 Results

[Text to write.]

5.1 Introduction

[Text to write.]

5.2 Experimental Data Overview

[Text to write.]

5.3 Command Detection Performance

[Text to write.]

5.4 Command Success and Missed Command Rates

[Text to write.]

5.5 False Activation Results

[Text to write.]

5.6 Command Detection and End-to-End Latency

[Text to write.]

5.7 Remote-Car Task Completion Results

[Text to write.]

5.8 Performance Variation Across Experimental Sessions

[Text to write.]

5.9 Optional Custom Machine Learning Results

[Text to write.]

5.10 Summary of Findings

[Text to write.]

6 Discussion

[Text to write.]

6.1 Introduction

[Text to write.]

6.2 Interpretation of Experimental Results

[Text to write.]

6.3 Evaluation Against Research Objectives

[Text to write.]

6.4 Reliability and Accuracy of Mental Commands

[Text to write.]

6.5 Latency and Real-Time Control Performance

[Text to write.]

6.6 Trade-Offs Between Reliability and Response Time

[Text to write.]

6.7 Implications for Consumer-Grade EEG Applications

[Text to write.]

6.8 Comparison with Previous Research

[Text to write.]

6.9 Limitations of the Framework and Experimental Setup

[Text to write.]

6.10 Future Improvements

[Text to write.]

6.11 Chapter Summary

[Text to write.]

7 Conclusion

[Text to write.]

7.1 Introduction

[Text to write.]

7.2 Summary of the Research

[Text to write.]

7.3 Main Contributions

[Text to write.]

7.4 Achievement of Research Objectives

[Text to write.]

7.5 Research Limitations

[Text to write.]

7.6 Recommendations for Future Research

[Text to write.]

7.7 Final Remarks

[Text to write.]

References

Besl, P. and McKay, N. D. (1992). A method for registration of 3-d shapes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 14(2):239–256.

Annex

A. Experimental Protocol

[Text to write.]

B. Participant Information and Consent Documents

[Text to write.]

C. Supplementary Experimental Results

[Text to write.]

D. System Architecture and Technical Configuration

[Text to write.]

E. Source Code and Repository Information

[Text to write.]

F. Experimental Data and Reproduction Instructions

[Text to write.]

Faculdade de Design,
Tecnologia e Comunicação
Universidade Europeia

2026

Campus de Santos . Av. D. Carlos I, 4,1200-649 Lisboa | Portugal
Telf: (+351) 213 030 600 . iade@iade.pt

MOD.IADE-315/V01-EN; 05-05-2023